

NanoGT Manual Scoring Sheet

Fill Parts A–C from Orphanet / OMIM / UniProt before looking at any precedent program.

A — Disease Identity

Disease name	
Orphanet ID	ORPHA:
OMIM phenotype ID	
Prevalence	
Inheritance (circle)	AR / XL-recessive / XL-dominant / AD / Mitochondrial / Unknown

Source: orpha.net → search disease name

B — Gene & Protein

Gene symbol	
CDS length (bp)	_____ bp (= amino acid length × 3)
Amino acid (aa) length	_____ aa
UniProt accession	
Protein class (circle)	Secreted / Lysosomal / Membrane / Intracellular
Subcellular location	
Is secreted?	Y / N
UniProt keywords	
GO terms	

Source: uniprot.org → search gene symbol → filter Human (reviewed) → Subcellular location + Keywords + Sequences tab

C — Tissues, HPO & Onset

Affected tissue(s) (circle all)	Liver / CNS / Retina / Muscle / Haematopoietic / Heart / Kidney
Key HPO terms	
Onset timing (circle)	Congenital/neonatal / Early childhood / Childhood / Adult / Chronic
Progressive neurodegeneration?	Y / N
Inferred pathway (circle)	Lysosomal storage / Coagulation / Motor neuron / Myopathy / Retinal / Amino acid metabolism / Urea cycle / Mitochondrial / Unknown

Source: Orphanet → Clinical signs & phenotypes → HPO terms. Onset from OMIM clinical synopsis.

C (continued) — Disease-Fixed Dimensions

Score once here, copy into every Part D

#	Dimension	Max	Scoring rule	Your score
8	Therapeutic Window	2.0	Adult/chronic → 2.0 · Progressive childhood → 1.5 · Early childhood → 1.2 · Neonatal → 0.8 · Congenital/fatal → 0.5	<u> </u> /2.0
9	Cross-Correction	1.0	Secreted lysosomal → 1.0 · Secreted → 1.0 · Non-secreted lysosomal → 0.8 · Intracellular/membrane → 0.2	<u> </u> /1.0
10	Immune Privilege	1.0	Retina 1.0 · CNS 0.9 · Liver 0.8 · Muscle/Heart 0.6 · Kidney 0.5 · Haematopoietic 0.3 (<i>best tissue wins</i>)	<u> </u> /1.0
11	Promoter Availability	1.0	Liver 1.0 (ApoE/hAAT, TBG) · Retina 1.0 (VMD2, GRK1) · CNS 0.8 · Muscle 0.8 · Haem 0.7 · Heart 0.6 · Kidney 0.4	<u> </u> /1.0
12	RoA Feasibility	1.0	Liver IV 1.0 · Muscle IV/IM 0.9 · Haem ex vivo 0.9 · Retina 0.8 · CNS intrathecal 0.7 · Heart 0.6 · Kidney 0.4	<u> </u> /1.0

D — Per-Program Scoring

Complete once for each precedent program. Copy dims 8–12 from Part C.

Precedent program name	
Vector	
Approval status (circle)	Approved / Phase 3 / Phase 2/3 / Phase 2 / Phase 1/2 / Phase 1
Designed for disease	
Vector cargo limit (bp)	_____ bp (AAV = 4,700 bp · LV = 8,000 bp)
Program CDS (bp)	_____ bp
Program protein class	Secreted / Lysosomal / Membrane / Intracellular
Program tissue target	
Program inheritance	AR / XL / Other
Program pathway	

#	Dimension	Max	Scoring rule	Score	Notes
1	Packaging Fit	2.0	Disease CDS ÷ vector cargo = utilisation % · >100% → FAIL (0.0) — stop, score nothing else ≤30% → 2.0 · 31–60% → 1.5 · 61–85% → 1.0 · 86–100% → 0.5	_____ /2.0	_____ _%
2	Tissue Tropism	2.0	Vector tropism AND program target both match → 2.0 Program target matches or vector covers ≥2 tissues → 1.5 Vector covers 1 tissue → 1.0 · No overlap → 0.3 · No data → 1.0	_____ /2.0	
3	Protein Class	2.0	Both lysosomal → 2.0 · Both secreted → 2.0 Both intracellular → 1.5 · Both membrane → 1.5 Partial extracellular → 1.0 · Mismatch → 0.5	_____ /2.0	
4	Inheritance	1.0	AR↔AR or XL↔XL → 1.0 · Any LOF↔LOF → 0.7 Unknown → 0.5 · Dominant/mito vs recessive → 0.3	_____ /1.0	
5	Pathway Similarity	2.0	Exact match → 2.0 · Related group → 1.5 · Different → 0.5 · Unknown → 1.0 Groups: <i>lysosomal_storage</i> · <i>coagulation</i> · <i>motor_neuron~myopathy</i> · <i>retinal</i> · <i>amino_acid-urea_cycle</i> · <i>mitochondrial</i>	_____ /2.0	
6	Approval Weight	1.0	Approved 1.0 · Ph3 0.8 · Ph2/3 or Withdrawn 0.7 · Ph2 0.6 · Ph1/2 0.5 · Ph1 0.4	_____ /1.0	
7	Immunogenicity (vector NAb)	2.0	Seroprevalence of vector: <10% → 2.0 · 10–19% → 1.5 · 20–39% → 1.0 · ≥40% → 0.5 AAV2 55% · AAV8 30% · AAV9 22% · AAV1 20% · AAV2/6 17% · AAVrh10 10% · AAV5 9% · LV 2%	_____ /2.0	sero= _____
8	Therapeutic Window	2.0	Copy from Part C	_____ /2.0	

9	Cross-Correction	1.0	<i>Copy from Part C</i>	<u> </u> /1.0	
10	Immune Privilege	1.0	<i>Copy from Part C</i>	<u> </u> /1.0	
11	Promoter Availability	1.0	<i>Copy from Part C</i>	<u> </u> /1.0	
12	RoA Feasibility	1.0	<i>Copy from Part C</i>	<u> </u> /1.0	

Score Calculation

Raw total (sum all 12 dimensions)	<u> </u> / 18.0
Composite score (raw ÷ 18 × 10)	<u> </u> / 10.0
Confidence (circle)	High (≥ 7.5) / Medium (5.0–7.4) / Low (< 5.0) / FAIL (packaging = 0)

My Interpretation

Which dimension drove the score highest / lowest?	
Code output matches? If not — which dim differs and why?	